

AI in Assistive Technology: Existing Standards to a New York Standard

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1 The Problem

Accessible interfaces do not guarantee dependable AI.

People with disabilities may rely on AI-enabled assistive technology (AT) to communicate, navigate, and carry out everyday tasks. A failure can affect safety, autonomy, and access.

Users carry too much of the burden

Assessing privacy, security, fairness, and safety product by product can add to the demands of daily life.

ILLUSTRATIVE EXAMPLE

A speech interface may meet accessibility guidelines while its AI still fails to recognize atypical speech.

2 What We Reviewed

18

instruments reviewed

2

review tiers

4

Tier 2 groups

We critically compared standards, laws, frameworks, and audit methods for their coverage of AI-enabled AT and their ability to demonstrate conformance.

Tier 1 · AI + disability

Two instruments directly address AI and disability; both come from outside the United States.

Tier 2 · Supporting instruments

Accessibility, AI governance, cybersecurity, and assurance provide complementary building blocks.

3 Tier 1: Strengths and Limitations

CAN-ASC-6.2:2025 · Canada

Contributes disability participation across the AI lifecycle, performance reporting by disability status, and plain-language explanations.

REVIEW FINDING

Broad AI scope and voluntary adoption leave a need for AT-specific tests, evidence, and a clear verification pathway.

NDIS / CSIRO framework · Australia

Co-designed for AI-enabled AT around user experience, value, quality, safety, privacy and security, and human rights.

REVIEW FINDING

Supports decisions about suitability, but does not establish measurable pass criteria or a conformance assessment mechanism.

4 Tier 2: Useful Parts, Incomplete Coverage

Each group contributes part of a standard. The review identifies a remaining gap in each.

Accessibility & nondiscrimination

WCAG 2.2; Section 508 / VPAT; ADA Title II; EEOC guidance; HHS Section 504 and Section 1557 rules
Contributes: Interface access and disability rights.

Gap: AI performance and most AT products remain outside these checks.

AI governance

EU AI Act; NY AI Act proposal; NY RAISE Act; NYC Local Law 144; NIST AI RMF; ISO/IEC 42001
Contributes: Risk management, oversight, and organizational accountability.

Gap: Disability and AT-specific evaluation are absent or limited.

Cybersecurity

NIST CSF 2.0; FDA §524B and device guidance
Contributes: Risk management and device security evidence.

Gap: Many consumer AT products fall outside medical-device oversight.

Audit & certification

IEEE CertifAIED; Lam et al., criterion audits
Contributes: Methods for independent verification.

Gap: Verification needs explicit disability and AT criteria.

5 What We Have Found So Far

01 Fragmented scope

No reviewed instrument combines AT coverage, disability-specific evaluation, and enforceable verification.

02 Missing operational criteria

Principles need test procedures, acceptable thresholds, and required vendor evidence.

03 Uneven accountability

Self-reporting and voluntary participation cannot substitute for independent assurance.

04 Security must remain accessible

Protection should not lock out the person who depends on the technology.

6 The Gap We Are Trying to Close

Move from broad principles to evidence that an AT product works for the people who depend on it.

Combine interface accessibility, AI performance, device security, and independent verification in one disability-centered standard.

7 What We Propose for New York

A comprehensive standard for AI-enabled assistive technology

PROPOSED REQUIREMENTS FROM THE REVIEW

Co-design throughout the lifecycle

Involve people with disabilities in defining needs, tests, acceptable risk, and ongoing review.

Test the interface and the AI

Use WCAG 2.2 Level AA as an interface baseline; require AI evaluation with disabled users.

Define measurable vendor evidence

Specify AT use cases, functional needs, test protocols, performance thresholds, and reporting.

Build in safety, privacy, and security

Cover consumer AT as well as medical devices; test accessible security controls and safe alternatives.

Require independent assurance

Audit against disability-specific criteria; connect conformance to procurement and accountability.

Monitor and improve

Collect accessible feedback, investigate failures, and reassess products after updates.

8 What's Next

1 Convene stakeholders

Bring together disabled users, caregivers, AT developers, service providers, and New York State agencies to define priorities and acceptable risk.

2 Develop AT-specific test criteria

Start with one category, such as speech recognition for atypical speech, and define use cases, measures, required evidence, and thresholds.

3 Pilot evaluations

Test real AI-enabled AT products for AI performance, accessibility, safety, privacy, and accessible security controls.

4 Draft a New York standard

Propose a New York standard with an independent conformance pathway linked to state procurement.

9 The Intended Outcome

Evidence users can trust. Accountability vendors can demonstrate.

Reduce the burden of evaluating every product and support safer, fairer, more independent use of AI-enabled AT.

Selected sources: [1] Accessibility Standards Canada, CAN-ASC-6.2:2025. [2] NDIA / CSIRO, AI-enabled AT Framework (2022). [3] W3C, WCAG 2.2. [4] NIST, AI RMF 1.0 and CSF 2.0. [5] ISO/IEC 42001:2023. [6] IEEE CertifAIED. [7] Lam et al., A Framework for Assurance Audits of Algorithmic Systems, FAcCT '24.